TC-Lab: what remains in the error?

Conductance variation, radiation and discrete measurements

Prepared for Alex Dowling

October 26, 2026

Work in progress. This layout draft contains audited intermediate results. Joint discrete-data fits, local information and all twenty canonical transfer folds are audited. Fifty-draw refitted recovery and initial/bin sensitivities are audited; remaining profiles and parameter-inclusive transfer are underway. It is not the completed study report.

Question	Evidence so far	Status
Can modest run variation help?	A 10% conductance band gives T1/T2 RMSE 0.605/0.118 °C in the joint bin-likelihood fit.	Audited; little nominal protocol-transfer gain.
Does radiation explain the rest?	Free radiation gives 0.590/0.119 °C in the joint fit; bin nll improves by 3.810.	Conditional change; boundary profile pending.
Are cold starts plausible?	Quantized hidden heater offsets remain below 0.515 °C, but exchange lag labels.	Fits audited; startup profile underway, equilibrium unverified.
Are measurements discrete?	All 21,614 temperatures match a shifted 0.3223 °C code grid.	Hardware evidence; threshold phase remains assumed.
Do lag labels identify physics?	At the loading boundary, a nominal forced-response lag swap changes T1 by at most 0.00430 °C.	Weak assignment; lowering the loading bound barely changes the mean.
Are residuals adequate for inference?	Deterministic code diagnostics retain input, phase and cross-channel structure.	Whitening unfinished; no IID claim.
Does the nominal fit transfer?	Omit one run: 1.750/0.754 °C; omit a protocol: 2.131/0.875 °C.	All canonical folds audited; parameter propagation pending.

1 Records and model

We accept the same-device ENV2 provenance as the study assumption. The twelve distinct records comprise nine sine tests (three repeats at each of 30, 120 and 300 s) and three step tests. Duplicate aliases do not count as experiments. Only within-run time is available; Git history does not establish recovery time between runs.

Both temperatures enter a coupled heater/sensor model. With $i = 1, 2$ and the other heater denoted by k,

$$C_H \dot{H}_i = U_{a,j}(A_j - H_i) + U_s(S_i - H_i) + U_x(H_k - H_i) + \beta z_i, \tag{1}$$

$$C_S \dot{S}_i = U_s(H_i - S_i), \tag{2}$$

$$L\dot{z}_1 = u + c\,u(1 - u/100) - z_1, \qquad z_2 = 0. \tag{3}$$

The recorded command is held from its timestamp to the next update. The empirical curvature is not a measured electrical power law. Each run has $A_j = T_{1,j}(0) + a_j$, with $|a_j| \le 2\,°\mathrm{C}$. The baseline starts $H_i(0) = S_i(0) = T_{i,j}(0)$ and $z_1(0) = 0$.

Absolute power calibration is unavailable. Estimate scale-free quantities

$$K_0 = \beta/U_{a,0}, \quad t_A = C_H/U_{a,0}, \quad r_S = C_S/U_s, \quad r = U_s/U_{a,0}, \quad q = U_x/U_{a,0}.$$

A constant shared heater calibration multiplier is absorbed into gain. Temperature-only fitting does not identify absolute conductances or heat capacities without an independent scale.

One physical coefficient varies across runs:

$$U_{a,j} = U_{a,0}e^{\delta_j}, \quad \sum_j \delta_j = 0, \quad \log(1-b) \le \delta_j \le \log(1+b).$$

Thus $K_j = K_0 e^{-\delta_j}$, $t_{A,j} = t_A e^{-\delta_j}$, $r_j = re^{-\delta_j}$ and $q_j = qe^{-\delta_j}$, while r_S stays shared. The hard band preserves coupling between experiments. It is not a population distribution or a fitted random-effects variance.

2 What the matched fits show

The old short-lag/curved-command coupled fit has T1 RMSE 1.022 °C. Orthogonal error projections attribute 74.6% of its T1 squared error to run offsets and another 19.3% to run trends. The additional fundamental sine projection contributes only 0.64%. These are descriptive projections, not independent physical causes. The earlier description of a one-degree amplitude error confused total RMSE with sine amplitude mismatch.

Gaussian comparison	T1 RMSE, °C	T2 RMSE, °C
Shared short lag and curved command	1.022	0.231
Conductance band 5%	0.646	0.147
Conductance band 10%	0.639	0.115
Conductance band 20%	0.641	0.115
10% band, affine ambient	0.728	0.111
10% band, hidden heater starts	0.615	0.111
10% convective band, free radiation	0.633	0.115

Widening the band adds little beyond 10% in this comparison. Two 10% deviations reach their lower bound; protocol-aligned ratios are fitted behavior, not measured airflow. Likelihood and unweighted RMSE need not rank models alike. These fits reuse all twelve records.

The earlier T1-only positive three-pole benchmark, with command curvature and affine ambient, reached 0.374 °C RMSE. Its different output set and ambient formulation prevent a matched likelihood or AIC comparison with these joint fits. It remains historical prediction context, without a unique physical assignment of its three poles.

Radiation scale

Use Kelvin in $Q_{\mathrm{rad}} = \epsilon\sigma A_{\mathrm{rad}}(H^4 - T_r^4)$. For illustrative area 0.0012 m^2, surroundings at 23 °C and surface temperature 70.54 °C, emissivity scenarios 0.05–0.9 give 0.021–0.383 W. Against a linear radiative loss through ambient, matched at 50 °C, the nonlinear remainder is only 0.0021–0.0370 W. A locally linear radiative contribution can already reside in fitted U_a.

Metal alone does not determine emissivity. Finish, effective area, view factors and surface temperature are unknown; a sensor reading is not a known heater-surface temperature. Published TC-Lab models include convection, radiation and sensor dynamics [1]. Their values supply modeling context, not calibration for these records.

The free-radiation fit refits convection and gain to avoid double counting. Its normalized differential radiation coefficient at 50 °C is 0.0344 relative to nominal

convection. Gaussian nll improves by 1.285. This is not a measured emissivity or a calibrated significance result.

Why assigning the two lags is difficult

At zero sensor loading, common/differential forced transfers factor as

$$G_{\pm}(s) = \frac{K}{(t_A s + 1 + \{0, 2q\})(1 + r_S s)(1 + Ls)}.$$

The output channels are $(G_+ + G_-)/2$ and $(G_+ - G_-)/2$. Exchanging r_S and L leaves these forced transfers unchanged in that limit.

At the fitted loading boundary $r = 0.005$, exchanging the nominal 14.98 s sensor lag and 2.69 s input lag changes the synthetic fast-sine T1/T2 waveforms by at most 0.00430/0.00185 °C. That calculation uses equal 20 °C starts and zero ambient offset. Unequal starts and ambient forcing can break the exchange. Repeated measurements can constrain sub-grid effects, so code spacing is not an absolute parameter-resolution limit. Lowering the positive loading bound from 0.005 to 0.0001 again reaches the bound and improves bin nll by only 0.01291; gain and fitted time scales barely change. This does not establish zero sensor heat capacity. Profiles must still test the lag assignments.

3 Discrete observations and residual checks

The hardware data, rather than just simulator output, are quantized. The inferred code values are round$(0.3223n - 50, 2)$. Their grid does not identify pre-quantization noise or exact thresholds.

There is also a temperature-reference freedom in the linear model. When cold initial sensor states and ambient can shift together, changing nearest to floor bins translates every interval and predicted temperature by half a code, 0.16115 °C, without changing the likelihood. An independent synthetic check verifies this identity with slack ambient bounds. Fixed initial states, active ambient limits or absolute-temperature radiation can break it. A threshold phase comparison must distinguish this reference shift from dynamic information.

For the unreported analogue temperature,

$$Y_i^* = m_i(\psi) + X_i + v_i, \qquad X_i = e^{-\Delta t_i/\ell} X_{i-1} + w_i,$$

with stationary variance S, process variance $S(1 - e^{-2\Delta t_i/\ell})$ and independent $v_i \sim N(0, R)$. The likelihood integrates conditional observation density over each code bin. Forward quadrature and backward smoothing account for correlated latent errors. Multiplying marginal bin probabilities gives a different likelihood.

Holding the 10% mean fit fixed, bin-likelihood refitting changes analogue measurement SD from about 0.094 to 0.0255 °C for T1, and from 0.072 to 0.0325 °C for T2. T2 correlation length changes from 7.9 to 54.8 s. These are conditional noise estimates, not calibration measurements. The joint 10% bin-likelihood fit is now independently audited. Its T1/T2 RMSE is 0.605/0.118 °C, with $K_0 = 0.8732$ °C/% and $t_A = 174.59$ s. Gain is within 0.1% of the matched Gaussian estimate. The fitted noise is

Joint bin fit	T1	T2
Latent correlated SD, °C	0.5362	0.0681
Correlation length, s	658.28	62.56
Analogue measurement SD, °C	0.0258	0.0325

The joint 5% and 20% bands are also audited, with RMSE 0.637/0.153 °C and 0.606/0.118 °C. Their nll values are 6279.4250 and 6238.2167, versus 6238.2987 for 10%. Four 5% deviations reach bounds; the wider band is not selected from training likelihood alone. The joint radiation fit gives 0.590/0.119 °C and nll 6234.4890. Its improvement is larger than in the Gaussian approximation, but is neither a calibrated significance result nor evidence of measured emissivity.

Fisher's score identity supplies derivatives from smoothed latent moments [2]. Selected points require independent physical propagation, Gaussian bin rectangles, grid/support convergence and score stationarity. A repaired rare-bin integration failure remains a numerical failure; it is not evidence against the shared physical model.

Whitening checks combine independent randomized predictive residuals [3], serial and cross-channel dependence, input/phase dependence, and nonrandomized down/same/up code probabilities. Fifty fixed-model simulations are complete with no failed draw. Across twenty independent randomizations, median maximum serial correlation over lags 1–60 is 0.0248/0.0245 for T1/T2; maximum cross-channel correlation has median 0.0224. These are comparable to the fixed-model references. T1's nonrandomized code-transition discrepancy has reference upper-tail rank 0.0588. However, deterministic conditional expected Gaussian-PIT scores expose structure that the added PIT jitter diluted:

Statistic	Data	Reference 97.5%
T1 input/score correlation magnitude	0.04429	0.02261
T1 maximum phase projection	0.02983	0.01971
T2 maximum ACF, lags 1–60	0.04695	0.04039
Cross-channel maximum, lags ±10	0.04895	0.02540

Each descriptive upper-tail rank is 1/51. These scores are discrete and not unit-variance Gaussian residuals. The same calculation is applied to all fifty references. Removing run trends and three sine harmonics leaves cross correlation 0.04719, above its matched reference 97.5% value 0.02603. Projection has not resolved the dependence or identified its cause. Fixed-model references do not refit parameters or repeat selection, so these ranks are not calibrated composite-null p-values. Whitening remains unfinished.

Updating the residual state with earlier reported codes reduces whole-data one-step reported-temperature RMSE to 0.103/0.087 °C, compared with physical free simulation at 0.605/0.118 °C. At a horizon of at least 60 s, it is 0.255/0.110 °C. Parameters come from the same twelve records. These are feedback-assisted in-sample predictions; they do not validate the free dynamics or establish independent innovations.

Continuous-Gaussian joint sensitivities give process-driving correlation 0.0908 or measurement correlation −0.00692 when estimated separately. Estimating both gives 0.11587/−0.02463 and improves Gaussian nll by 6.14891. All have independent dense-covariance audits. This is a comparison on discrete temperatures using a continuous likelihood; it does not establish calibrated joint-bin probabilities or an environmental mechanism.

4 Uncertainty, transfer and open questions

The independently audited information matrix distinguishes the centered tangent space from the selected active constraint face. Conditional local standard errors are

Coordinate	IID analogue	Correlated analogue	Joint bin
$\log K$	0.0007263	0.0033010	0.0033620
$\log t_A$	0.0005152	0.0030056	0.0030888
$\log L$	0.0479381	0.0309354	0.0324303

The first two columns hold covariance known and use identical analogue marginal variance. Correlation increases gain and thermal-time SEs by 4.55 and 5.83, while reducing the short-lag SE to 0.645 of its IID value. The bin column also estimates noise nuisances; its difference does not isolate quantization. These are selected-basin, active-face approximations, not calibrated intervals.

T1 log latent SD and log correlation length have local correlation 0.9845. Conditioning the residual process on the initial code changes their estimates from 0.5362 °C/658.3 s to 0.8763 °C/1761.9 s, while $2S/\ell$ changes only 0.22% and the mean parameters barely change. The data constrain the diffusion rate more clearly than its stationary variance/length split.

Allowing each cold heater/sensor pair within its first reported bin adds 24 bounded coordinates. Its audited nll falls to 6209.5536, gain to 0.8697 °C/% and short lag to 2.72054 s; several initial states reach edges. T1 free-simulation RMSE increases to 0.655 °C, while T2 falls to 0.113 °C. Better correlated-bin likelihood does not imply better noise-free simulation. These bin bounds do not integrate an initial measurement-noise posterior.

An observation-space check prevents a bin-reference shift from masquerading as improved prediction. The freely initialized nearest/floor fits differ in expected reported output by less than 0.00030 °C. Exact half-grid translation preserves those expectations within 3.1×10^{-12} °C. Their different raw code-minus-analogue-reference RMSE therefore does not demonstrate different predictive quality. The nearest free-initial fit's reported-code RMSE is 0.655/0.108 °C; stationary analogue noise and code integration are included.

Hidden-heater-only ±2 and ±5 °C sensitivities coincide, with maximum selected offset 0.514 °C. They favor sensor time 2.667 s and input lag 14.898 s, reversing the baseline labels. Their reported-code RMSE is 0.682/0.109 °C, against baseline 0.605/0.114 °C. A better likelihood did not improve T1 free prediction. The representative startup profile is pending. These changes are distinct from the earlier Gaussian hidden-start comparison.

Profiles refit all ambient, conductance and noise nuisances. Sensor-loading bounds, initial-state sensitivities and lag alternatives make routine Wald intervals unsafe without those checks.

The nine-point lag grid passed independent audits. Twice the nll increase is 552.445 at $L = 7$ s but falls to 6.253 at 15 s, where the fitted sensor time is 2.643 s. A further audited fixed point at $L = 14.79006$ s fits sensor time 2.7314 s and gives twice the nll increase 4.78730. Both starts converge, without improving the reference. These checks expose a second region; they do not locate its free minimum, interpolate endpoints or calibrate an exclusion of that lag assignment.

Fifty refitted recovery collections

All fifty declared generated collections and all one hundred free/true-gain point audits passed, with no missing outcome. The nominal 95% gain likelihood-ratio region includes the generating value in 49/50 collections. Its independently inverted exact-binomial 95% Monte Carlo interval is 89.35–99.95%. This is conditional on the generating model, initialization, bins, bounds and fitting algorithm. It does not establish real-device coverage.

Fifteen refits exchange the sensor/input lag labels. Unordered fast and slow 2.5–97.5% repeated-fit ranges are 2.547–2.898 s and 14.419–15.576 s. These are estimator quantiles, not physical confidence intervals. Gain's sampling SD is 0.898 times

its local SE; log thermal time spreads 4.780 times farther than the selected-face approximation. Named lag-coordinate spreads mix the two label regions and should not be summarized as one Gaussian error.

Refitting preserves the residual inadequacy finding. The reference 97.5% values for T1 input, T1 phase, T2 ACF and cross-channel diagnostics become 0.01689, 0.02146, 0.03894 and 0.03014, respectively. The hardware values in the earlier table exceed every one of the fifty refitted references. These are descriptive selected-diagnostic ranks, without multiple-testing or structure-selection calibration. Randomized PIT agreement has not established approximately IID innovations.

All twenty canonical transfer folds

Twelve fits omit one run each. Four band-model fits and four shared-conductance fits omit one protocol each (30, 120 or 300 s sine, or step). Every selected fit and canonical forecast passed independent audits. No later test temperatures estimate ambient, conductance or physical parameters; only the first temperature pair initializes the physical states. Each record appears once per comparison group, 10,795 scored readings per channel. Groups reuse these same observations; all records informed structure selection, so transfer is retrospective.

Nominal reported-mean RMSE, °C	T1	T2
Band: omit one run	1.750	0.754
Band: omit protocol	2.131	0.875
Shared: omit protocol	2.171	0.894

Allowing conductance variation improves the all-data fit much more than these nominal protocol predictions. Other repeats in training help the leave-run case. Run mean errors account for 84.9/72.5% of its T1/T2 SSE; removing them leaves RMSE 0.680/0.395 °C. For omitted protocols, the band model gives 85.7/73.2% and 0.806/0.453 °C. Offset removal uses the test outcomes and cannot serve as a blind correction.

Range and group	Coverage, % T1/T2	Mean width, °C T1/T2
Band/run, noise	46.01/47.56	2.093/0.291
Band/run, conditions + noise	100.00/99.94	9.160/5.170
Band/protocol, noise	32.07/36.66	2.010/0.303
Band/protocol, conditions + noise	99.57/99.95	9.072/5.173
Shared/protocol, noise	50.25/53.46	3.606/0.811
Shared/protocol, conditions + noise	94.01/99.81	6.769/3.969

Nominal conditions set the unknown run's conductance ratio to one and ambient offset to zero. Condition ranges allow ±2 °C ambient and, for the band model, ±10% conductance. They combine hard scenario limits with marginal noise quantiles. Higher coverage accompanies broad ranges; neither candidate has established calibrated total-95% predictions or an environmental cause. The shared model's larger residual variance gives broader noise-only ranges. Correlated readings do not become independent validation replicates.

The first complete parameter propagation check omits the three fast-sine runs. Its audited, box-truncated local normal fixes the selected active loading bound and conditions on one likelihood basin. Adding 256 draws changes condition-plus-noise mean widths from 9.119/5.134 °C to 9.223/5.206 °C. Using 256 instead of the shared-prefix 64 draws moves 2.26/2.69% of endpoints, by at most one reported code (0.33 C). This checks mixture resolution, not coverage calibration or released-bound/basin uncertainty. Four full components are audited; sixteen remain, including two stopped information gates under investigation. The local joint-bin log-input-lag SE without fast-sine data is 0.1232, versus 0.0324 in the twelve-run fit.

Each retained subset also defines a different geometric nominal conductance. Com-

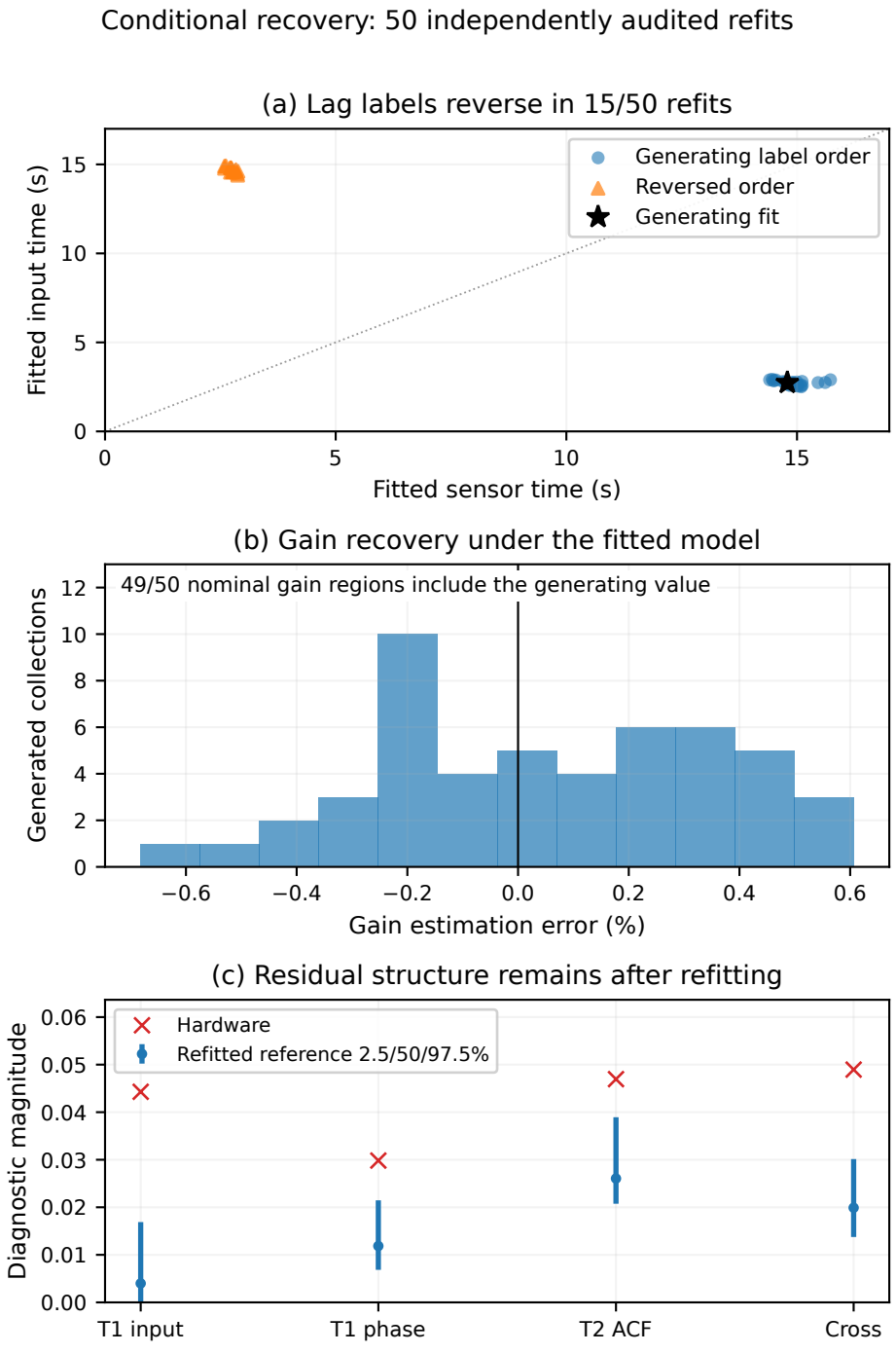


Figure 1: Conditional recovery from fifty fully audited refits. Lag labels exchange while the two time scales remain separated. The gain calculation checks nominal scalar LR coverage only. Hardware residual diagnostics exceed the refitted ranges; the analysis has not resolved their cause.

pare invariant combinations $K_0/t_A = \beta/C_H$, $r/t_A = U_s/C_H$ and $q/t_A = U_x/C_H$ before treating shifts in K_0 as physical change. Centering a hypothetical twelve-run cohort with the same hard band permits an omitted run's ratio to a nine-run training nominal in $[0.86894, 1.13551]$. A source-bound arithmetic check verifies these limits and unchanged canonical forecasts. A separately declared forecast stress check will propagate this range while preserving the training fits. It is not a transported cohort likelihood or a corrected coverage calibration.

The refitted checks retain input-linked residual structure. Remaining questions include whether the conductance band transfers across protocols, how startup/threshold assumptions assign the lag labels, and which uncertainty statements survive refitting and simulation. A future hood experiment should log ambient temperature, absolute time, prior run recovery, airflow settings, firmware and electrical power. A hood can add forced or fluctuating convection; it does not automatically supply a constant environment. No new hardware or DoE is part of this study.

Evidence and status

This draft uses terminal, independently audited intermediate artifacts in the committed remaining-error study. The full decision-gate audit, final figures and parameter-inclusive uncertainty/transfer results are pending. It will be replaced after those gates are assessed.

References

[1] Park, Martin, Kelly and Hedengren (2020). Benchmark temperature microcontroller for process dynamics and control. *Computers & Chemical Engineering*, article 106736. https://doi.org/10.1016/j.compchemeng.2020.106736.

[2] Poyiadjis, Doucet and Singh (2011). Particle approximations of the score and observed information matrix in state space models with application to parameter estimation. *Biometrika*, 98, 65–80. https://www.stats.ox.ac.uk/~doucet/poyiadjis_doucet_singh_particlescoreparameterestimation.pdf.

[3] Dunn and Smyth (1996). Randomized quantile residuals. *Journal of Computational and Graphical Statistics*, 5, 236–244. https://gksmyth.github.io/pubs/residual.pdf.